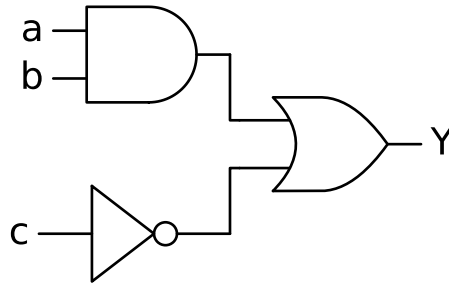


Boolean Algebra Worksheet

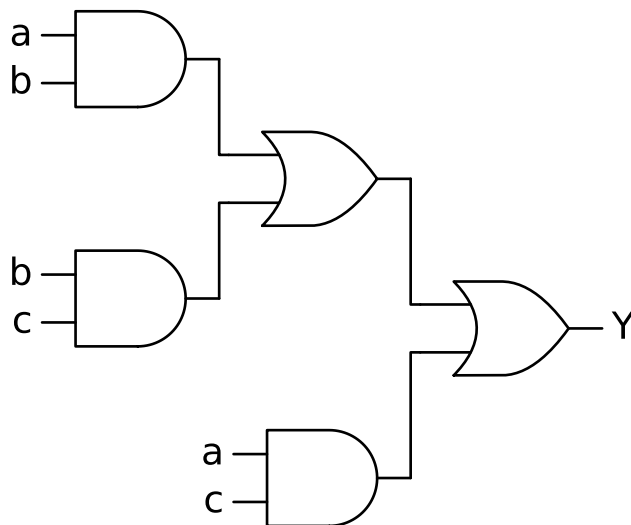
Reminder: $A \oplus B = \overline{A}B + A\overline{B}$ (XOR)

Problem 1.



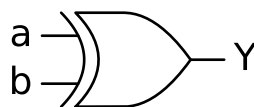
Write the Boolean expression for output Y, then draw its truth table.

Problem 2.



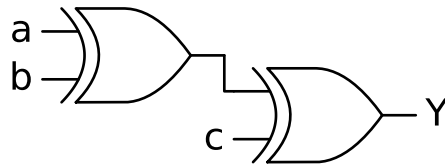
Write the Boolean expression for output Y, then draw its truth table.

Problem 3.



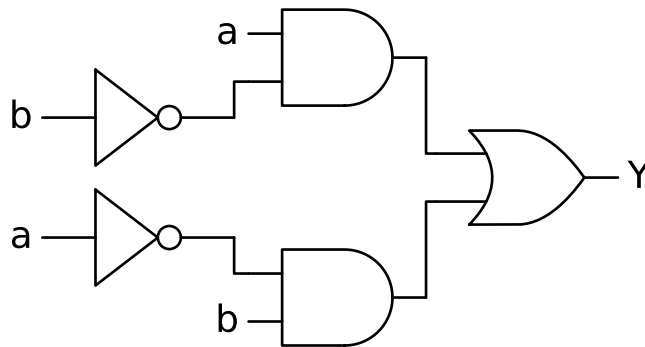
Write the Boolean expression for output Y, then draw its truth table.

Problem 4.



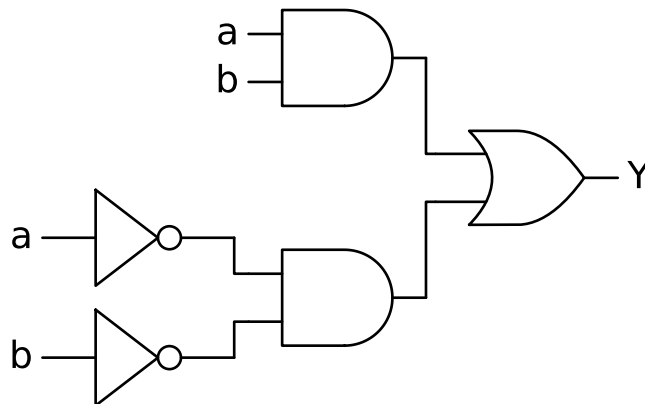
Write the Boolean expression for output Y , then draw its truth table.

Problem 5.



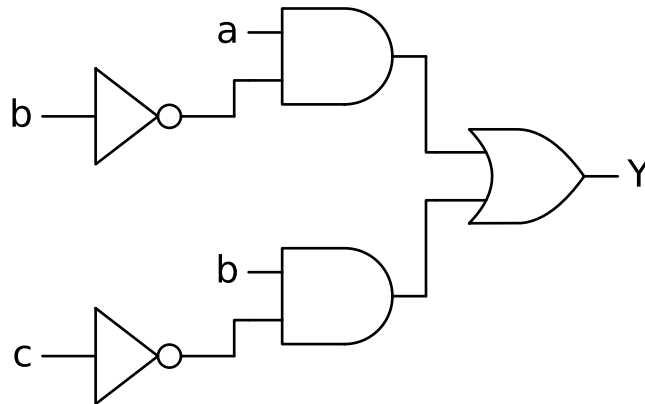
Write the Boolean expression for output Y , then draw its truth table.

Problem 6.



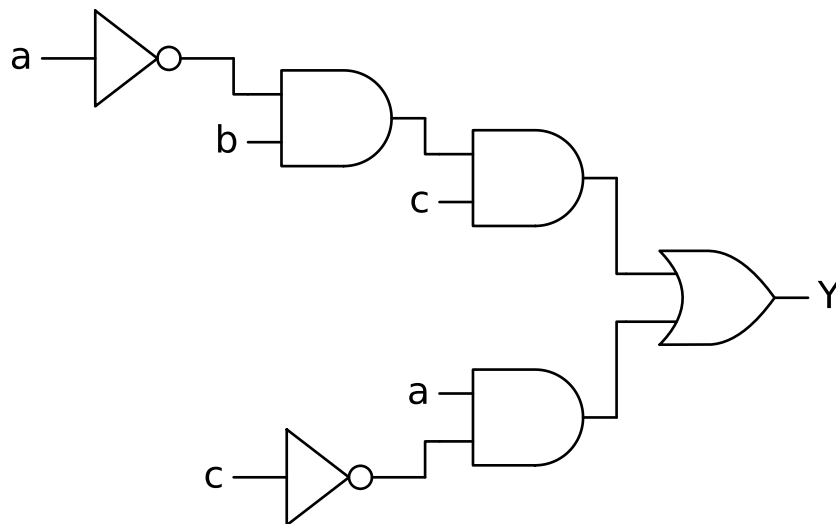
Write the Boolean expression for output Y , then draw its truth table.

Problem 7.



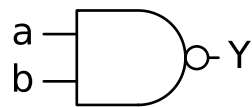
Write the Boolean expression for output Y , then draw its truth table.

Problem 8.



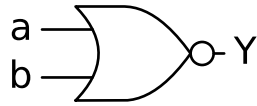
Write the Boolean expression for output Y , then draw its truth table.

Problem 9.



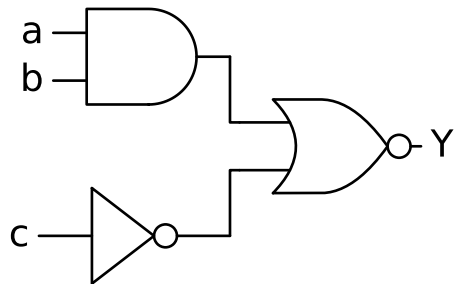
Write the Boolean expression for output Y , then draw its truth table.

Problem 10.



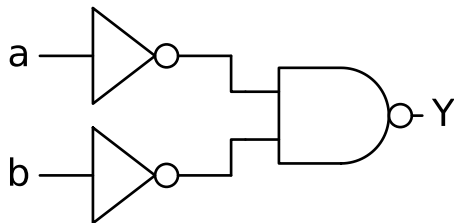
Write the Boolean expression for output Y , then draw its truth table.

Problem 11.



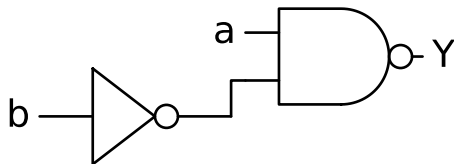
Write the Boolean expression for output Y , then draw its truth table.

Problem 12.



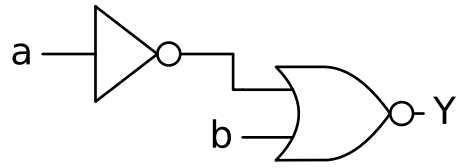
Write the Boolean expression for output Y , then draw its truth table.

Problem 13.



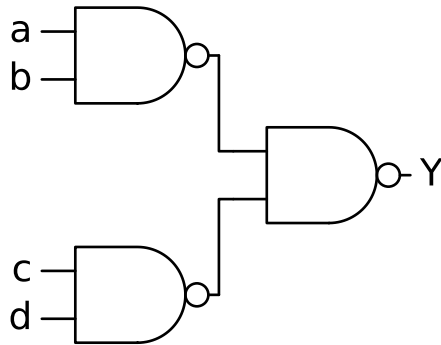
Write the Boolean expression for output Y , then draw its truth table.

Problem 14.



Write the Boolean expression for output Y , then draw its truth table.

Problem 15.



Write the Boolean expression for output Y , then draw its truth table.

Problem 16.

Draw the logic-gate circuit for:

$$Y = A \cdot (B + C)$$

Problem 17.

Draw the logic-gate circuit for:

$$Y = \overline{A \cdot B} + C$$

Problem 18.

Draw the logic-gate circuit for:

$$Y = (A + B) \cdot (A + C)$$

Problem 19.

Draw the logic-gate circuit for:

$$Y = A \oplus B \cdot C$$

Problem 20.

Draw the truth table for:

$$Y = A \cdot B + C$$

Problem 21.

Draw the truth table for:

$$Y = A \oplus B \oplus C$$

Problem 22.

Draw the truth table for:

$$Y = \overline{A + B} \cdot C$$

Problem 23.

Draw the truth table for:

$$Y = A \cdot \overline{B} + \overline{A} \cdot C$$
